2025 AMC 10A Problem 3 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 3. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 3

Problem:
How many isosceles triangles are there with positive area whose side lengths are all positive integers and whose longest side has length 2025?

Answer Choices:

A. 2025

B. 2026

C. 3012

D. 3037

E. 4050

Solution:

We want isosceles triangles with integer sides and longest side 2025. Let the sides be a, a, b.

Case 1: The equal sides are longest.

Then

$$a = 2025, \quad b \leq 2025.$$

Since b is a positive integer and $b \leq 2025$, we have

$$b = 1, 2, \ldots, 2025,$$

giving 2025 triangles.

Case 2: The base is longest.

Now

$$b = 2025, \quad a < 2025.$$

By the triangle inequality

$$2a > b \quad \Rightarrow \quad 2a > 2025 \quad \Rightarrow \quad a > \frac{2025}{2} = 1012.5.$$

So

$$a = 1013, 1014, \ldots, 2024,$$

which gives us $2024 - 1013 + 1 = 1012$ values of a, and thus 1012 triangles.

Combining our cases, we get a total of

$$2025 + 1012 = \boxed{\textbf{(D) } 3037}$$